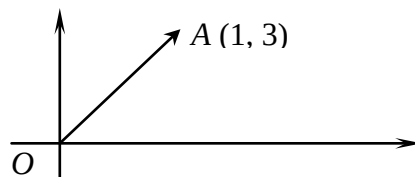


Question 1.

Marks

- (a) Let $z_1 = 2 - i$ and $z_2 = iz_1$, find $z_1 - z_2$. 2
- (b) Let $z = \frac{-9 + 15i}{4 - i}$,
- (i) By simplifying $\frac{-9 + 15i}{4 - i}$, express z in the form $a + bi$, 3
where a and b are real numbers.
- (ii) Hence, or otherwise find $|z|$ and $\arg z$. 2
- (c) Solve $z^2 = 35 - 12i$, expressing your answer in the form $a + bi$. 3
- (d) Sketch the region on an Argand diagram neatly, where the inequalities 4
 $|z - 2| \leq \operatorname{Re}(z)$ and $\operatorname{Im}(z) \geq 0$ hold simultaneously.
- (e) Let point A , below, represent the complex number $1 + 3i$ in the Argand diagram. 2



Not to scale

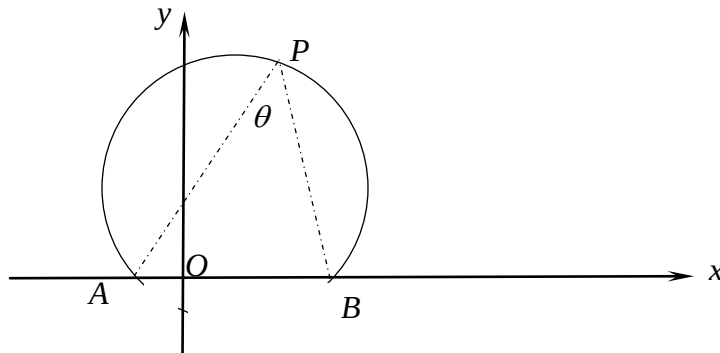
OA is rotated 45° anticlockwise about O and enlarged by a factor of $\sqrt{2}$.
Find this new complex number in the form $a + bi$.

Question 2. [Start a New Page]

- (a) Given $1 - 2i$ is a root of $x^2 + bx + c = 0$, where b and c are real numbers, 3
Find the value of bc .
- (b) (i) Sketch neatly on the same diagram the loci of z when: 2
(α) $|z - (3 + 2i)| = 2$. (β) $|z + 3| = |z - 5|$.
[clearly label each sketch]
- (ii) Hence write down all the values of z which satisfy simultaneously: 2
 $|z - (3 + 2i)| = 2$ and $|z + 3| = |z - 5|$.
- (iii) Determine the values of k for which the simultaneous equations: 2
 $|z - (3 + 2i)| = 2$ and $|z - 2i| = k$ have exactly one solution for z .
- (iv) For the locus of z when $|z - (3 + 2i)| = 2$, 2
find the maximum value of $\arg z$.

- (c) The diagram below shows the locus for all complex numbers z in the Argand diagram, such that $\arg\left(\frac{z-3}{z+1}\right) = \frac{\pi}{3}$.

The angle formed by the intervals AP and BP is θ , where P represents the complex number z and A is $(-1, 0)$ and B is $(3, 0)$.



Not to scale

- (i) Copy this diagram neatly and find θ , giving reasons. 2
- (ii) Find the coordinates for the centre of this circle. 2

Question 3. [Start a New Page]

- (a) Let P , Q and R represent the complex numbers z_1 , z_2 and z_3 respectively. 3
By drawing a diagram or otherwise, determine the geometric properties that characterize $\triangle PQR$ if $z_2 - z_1 = i(z_3 - z_1)$?
Give clear reasons for your answer.

- (b) Given $z = \cos \theta + i \sin \theta$ and that m and n are integers.

(i) Show that $\frac{z^n + z^{-n}}{2} = \cos n\theta$. 2

- (ii) Hence, using part (i), show that 2

$$\cos m\theta \times \cos n\theta = \frac{1}{2} [\cos(m+n)\theta + \cos(m-n)\theta]$$

- (c) Let $P(z) = z^8 - \frac{5}{2}z^4 + 1$. The complex number α is a root of $P(z) = 0$.

(i) Show that $i\alpha$ and $\frac{1}{\alpha}$ are also roots of $P(z) = 0$. 3

- (ii) Find one of the roots of $P(z) = 0$ in exact form. 3

- (iii) Hence find all the roots of $P(z) = 0$. 2

Question 4. [Start a New Page]	Marks
(a) The point P on the Argand diagram represents the complex number z , where z satisfies $\frac{1}{z} + \frac{1}{\bar{z}} = 1$. Find the locus of P as z varies	3
(b) (i) State the triangle inequality for the two complex numbers z_1 and z_2 .	1
(ii) If $z_1 = 3 + 4i$ and $ z_2 = 13$, find the greatest value for $ z_1 + z_2 $.	1
(iii) Further, if $0 < \text{Arg } z_2 < \frac{\pi}{2}$ and $ z_1 + z_2 $ has this greatest value, find z_2 .	2
(c) Given $w = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$,	
(i) Show that w^k is a solution of $z^5 - 1 = 0$, where k is an integer.	2
(ii) Explain why $1 + w + w^2 + w^3 + w^4 = 0$.	2
(iii) Hence, or otherwise find the value of: $(w-1)(1 + 2w + 3w^2 + 4w^3 + 5w^4)$	2
(iv) Given that $S = \frac{w}{1-w^2} + \frac{w^2}{1-w^4} + \frac{w^3}{1-w} + \frac{w^4}{1-w^3}$, Show that $S = 0$.	2

THE END

MATH. EXT 2 SOLUTIONS, TERM 4 2007

MATHEMATICS Extension 2: Question 1

Suggested Solutions

Marks

Marker's Comments

(a) $z_1 - z_2 = 2 - i - i(2 - i)$
 $= 2 - i - 2i - 1$
 $= 1 - 3i$

[2]

(b) (i) $z = \frac{(-9 + 15i) \times (4 + i)}{(4 - i)(4 + i)} = \frac{-36 - 9i + 60i - 15}{17}$

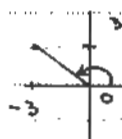
[3]

$z = \frac{-51 + 51i}{17} = -3 + 3i$

(ii) $|z| = \sqrt{9 + 9} = \sqrt{18} = 3\sqrt{2}$

$\arg z = \frac{3\pi}{4}$

[2]



1/2 For $10 + i$, 1/2 For 17
 1 For expanding
 1 For $a = -3$
 1 For $b = 3$

1 For $|z| = \sqrt{18} = 3\sqrt{2}$
 1 For $\arg z = \frac{3\pi}{4}$

(c) Let $z = x + iy$, $x, y \in \mathbb{R}$
 $z^2 = x^2 - y^2 + 2ixy = 35 - 12i$

METHOD 1

$x^2 - y^2 = 35$ — (1)

$xy = -6$ — (2)

so $x^2y^2 = 36$

and $x^4 - x^2y^2 = 35x^2$

i.e. $x^4 - 36 = 35x^2$

$x^4 - 35x^2 - 36 = 0$

$(x^2 - 36)(x^2 + 1) = 0$

$x = \pm 6$ or no real solutions

$\therefore y = \mp 1$

$\therefore z = 6 - i$ or $-6 + i$

[3]

1 For each solution

1 For $6 - i$

1 For $-6 + i$

(d) Let $z = x + iy$

$\operatorname{Re}(z) = x$

$|z - z| = \sqrt{(x - x)^2 + y^2}$

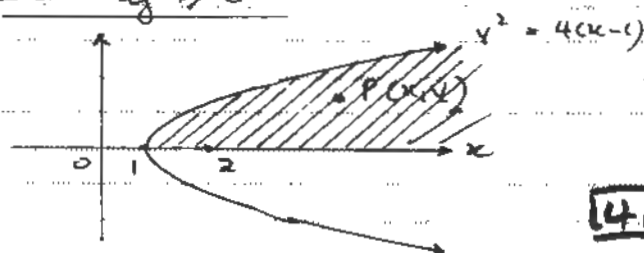
$\operatorname{Im}(z) = y$

$\therefore \sqrt{(x - x)^2 + y^2} \leq x$

$x^2 - 4x + 4 + y^2 \leq x^2$

$y^2 \leq 4(x - 1)$

and $xy \geq 0$



[4]

1/2 each $\operatorname{Re}(z) = x$
 $\operatorname{Im}(z) = y$

1 For $\sqrt{(x-x)^2 + y^2} \leq x$

1 For $y^2 \leq 4(x-1)$

1/2 For sketch of

1 For shaded region

(e) New number $= (1 + 3i) \times \sqrt{2} \times \cos \frac{\pi}{4}$
 $= (1 + 3i) \times \sqrt{2} \times \frac{1}{\sqrt{2}} (1 + i)$

[2]

$= -2 + 4i$

MATHEMATICS Extension 2: Question 2

Suggested Solutions

Marks

Marker's Comments

(a) $(1-z)$ is given.
so $(1+z)$ is also a root as all the coefficients are real numbers.

1 mk

$$\alpha\beta = \frac{c}{a} = c = (1+z)(1-z) = 5$$

1/2

$$\alpha + \beta = -\frac{b}{a} = -b = 1+2i + 1-2i$$

$$-b = 2$$

$$b = -2$$

1/2

$$\text{so } bc = -2 \times 5 = -10$$

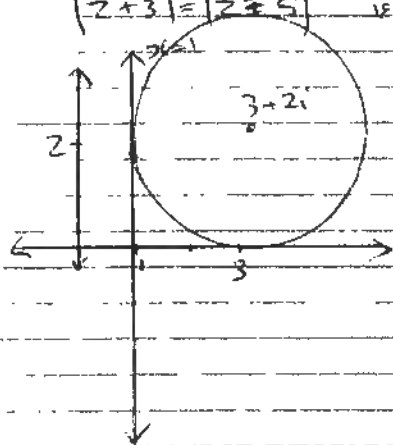
1 mk

(b)(i) $|z - (3+2i)| = 2$ is a circle centre $(3+2i)$ radius 2

1/2

$|z+3| = |z-5|$ is the perp. bisector of interval joining -3 and 5 , i.e. $x=1$

1/2 for $x=1$



1 for circle
ind. centre

(ii) since $x=1$ is a tangent to the circle there's only one point of intersection
 $z = 1+2i$

1 for answer
1 for explanation

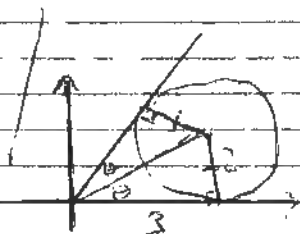
(iii) $|z-2i|=K$ represents a circle, centre $2i$, radius K .

1 mk for each
answer



is only one solution for z when $K=1$ or $K=5$

(iv) max arg $z = 2\theta$
 $= 2 \tan^{-1}(2/3)$
 $= 1.1071487$
 (63.43°)



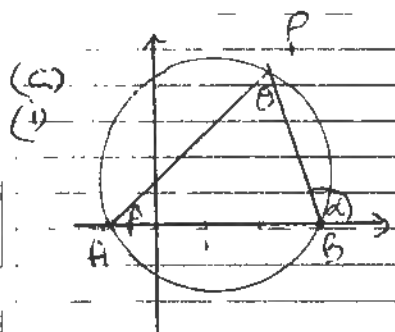
1/2 for diagram/explanation
1 for $\theta = \tan^{-1}(2/3)$
1/2 for answer

MATHEMATICS Extension 2: Question 2

Suggested Solutions

Marks

Marker's Comments



$$\arg(z-3) = \alpha$$

$$\arg(z+1) = \beta$$

$$\alpha - \beta = \theta$$

(exterior angle
of a triangle
equals the sum of
2 interior opposite angles)

$$\text{but } \alpha - \beta = \frac{\pi}{3}$$

$$\therefore \theta = \frac{\pi}{3}$$

(ii) centre will lie on $x=1$, since the perpendicular bisector of the chord AB passes through the centre



$$\angle AOB = 2\theta$$

$$\angle AOB = \frac{2\pi}{3}$$

$$\frac{y}{2} = \tan \frac{\pi}{6}$$

(angle at centre is
double the angle at
the circumference)

$$y = \frac{2}{3}\sqrt{3}$$

$\therefore$ centre is $(1, \frac{2}{3}\sqrt{3})$

1st mark

2nd mark

1st mark

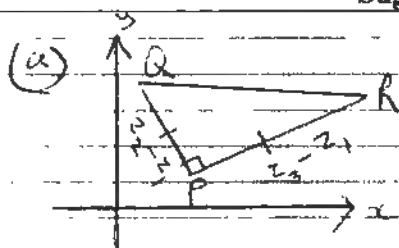
2nd mark

MATHEMATICS Extension 2: Question...3

Suggested Solutions

Marks

Marker's Comments



$\triangle PQR$ is right angled at $P(z_1)$ and is isosceles
 $PQ = PR$.

This is because the multiplication of $(z_3 - z_1)$ by i rotates this length 90° anticlockwise, it doesn't change the length as $|z_2 - z_1| = |i(z_3 - z_1)|$
 $= |i| \times |z_3 - z_1|$
 $= |z_3 - z_1|$

1 for answer/diagram

1

1

$$(b) \frac{z^n + z^{-n}}{2} = \frac{(e^{i\theta})^n + (e^{-i\theta})^n}{2}$$

$$= \frac{\cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta)}{2}$$

$$= \frac{\cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta}{2}$$

$$= \frac{2 \cos n\theta}{2}$$

$$= \cos n\theta$$

QED

1 mk

1 mk

$$(ii) \cos(m\theta) \times \cos(n\theta) = \frac{(z^m + z^{-m})}{2} \times \frac{(z^n + z^{-n})}{2}$$

$$= \frac{z^{m+n} + z^{-m-n} + z^{m-n} + z^{-m+n}}{4}$$

$$= \frac{z^{m+n} + z^{-(m+n)} + z^{m-n} + z^{-(m-n)}}{4}$$

$$= \frac{2 \cos(m+n)\theta + 2 \cos(m-n)\theta}{4}$$

$$= \frac{1}{2} [\cos(m+n)\theta + \cos(m-n)\theta]$$

QED

1/2

1/2

1/2

1/2

MATHEMATICS Extension 2: Question 3

Suggested Solutions

Marks

Marker's Comments

$$(c)(i) \quad P(z) = z^8 - \frac{5}{2}z^4 + 1$$

α is a root of $P(z) = 0$

$$\therefore \alpha^8 - \frac{5}{2}\alpha^4 + 1 = 0$$

$$P(i\alpha) = (i\alpha)^8 - \frac{5}{2}(i\alpha)^4 + 1$$

$$= i^8 \alpha^8 - \frac{5}{2}(i^4 \alpha^4) + 1$$

$$= \alpha^8 - \frac{5}{2}\alpha^4 + 1$$

$$= 0$$

$$P\left(\frac{1}{\alpha}\right) = \left(\frac{1}{\alpha}\right)^8 - \frac{5}{2}\left(\frac{1}{\alpha}\right)^4 + 1$$

$$= \frac{1}{\alpha^8} \left(1 - \frac{5}{2}\alpha^4 + \alpha^8\right)$$

$$= \frac{1}{\alpha^8} \times 0$$

$$= 0$$

$\therefore i\alpha$ and $\frac{1}{\alpha}$ are also roots of $P(z)$

$$(ii) \quad z^8 - \frac{5}{2}z^4 + 1 = 0$$

$$2z^8 - 5z^4 + 2 = 0$$

$$(2z^4 - 1)(z^4 - 2) = 0$$

$$\therefore z^4 = \frac{1}{2} \text{ or } z^4 = 2$$

one root is $z = \sqrt[4]{2}$

(iii) the roots are $\pm\sqrt[4]{2}, \pm i\sqrt[4]{2}, \pm\frac{1}{\sqrt[4]{2}}, \pm\frac{i}{\sqrt[4]{2}}$

2 marks

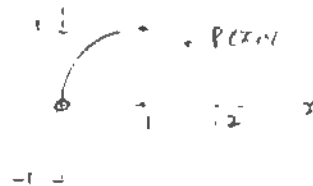
MATHEMATICS Extension 2: Question 4

Suggested Solutions

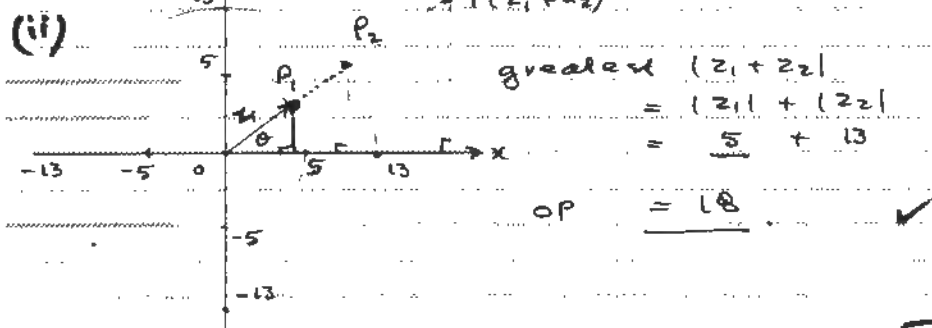
Marks

Marker's Comments

(a) Let $z = x + iy$ so $\frac{1}{z} + \frac{1}{\bar{z}} = 1$ [3]
 $\bar{z} = x - iy$
 $\frac{z + \bar{z}}{z\bar{z}} = 1$
 $\frac{2x}{x^2 + y^2} = 1$
 $\therefore x^2 + y^2 = 2x$
 $\therefore x^2 - 2x + y^2 = 0$
 $\therefore (x-1)^2 + y^2 = 1$
 $\therefore$ locus of z is $(x-1)^2 + y^2 = 1$ except $(0,0)$
a circle $c(1,0)$
 $r=1$



(b) (i) $|z_1 + z_2| \leq |z_1| + |z_2|$ ✓ equality iff collinear ($z_2 = kz_1$)



(iii) As z_1 and z_2 are collinear [2]
 $OP = 18$
 $\therefore \vec{OP}_2 = z_2 = \frac{13}{5} \vec{OP}_1 = \frac{13}{5} z_1 = \frac{13}{5} (3+4i)$

$z_2 = 13 \operatorname{cis} \theta$
where $\tan \theta = \frac{4}{3}$

(c) (i) $w = \operatorname{cis} \frac{2\pi}{5}$
 $[w^k = \operatorname{cis} \frac{2\pi k}{5} \text{ de Moivre's Thm}]$
 $\therefore z^5 - 1 = (w^k)^5 - 1 = (w^5)^k - 1$
 $= (\operatorname{cis} \frac{2\pi \times 5}{5})^k - 1$ [2]
 $= (\operatorname{cis} 2\pi)^k - 1$
 $= 1 - 1$
 $\therefore z^5 - 1 = 0$
 $\therefore w^k$ is a solution of $z^5 - 1 = 0$ $k \in \mathbb{Z}$

(ii) As $z^5 - 1 = 0$ $\therefore$ 5 roots $= w^0, w^1, w^2, w^3, w^4$
 $= 1, w, w^2, w^3, w^4$
and $\sum \alpha = 1 + w + w^2 + w^3 + w^4 = -\frac{1}{2} = 0$

[2]

MATHEMATICS Extension 2: Question 4.

Suggested Solutions

Marks

Marker's Comments

(c) (iii)

[2]

$$(w-1)(1+2w+3w^2+4w^3+5w^4)$$

$$= w + 2w^2 + 3w^3 + 4w^4 + 5 \times 1 - 1 - 2w - 3w^2 - 4w^3 - 5w^4$$

$$= -w - w^2 - w^3 - w^4 + 5 - 1$$

$$= -(w + w^2 + w^3 + w^4) + 4$$

$$= -(-1) + 4$$

$$= 5$$

(iv) METHOD 1: $S = \frac{w}{1-w^2} + \frac{w^2}{1-w^4} + \frac{w^3}{1-w} + \frac{w^4}{1-w^3}$

$$\therefore wS = \frac{w^2}{1-w^2} + \frac{w^3}{1-w^4} + \frac{w^4}{1-w} + \frac{1}{1-w^3}$$

$$= \frac{w^2}{1-w^2} + \frac{w^4}{w-1} + \frac{w^4}{1-w} + \frac{w^2}{w^2-1}$$

[2] $= \frac{w^2}{1-w^2} + \frac{w^2}{w^2-1} + \frac{w^4}{w-1} + \frac{w^4}{1-w}$

$$= \frac{w^2}{1-w^2} - \frac{w^2}{1-w^2} + \frac{w^4}{w-1} - \frac{w^4}{w-1}$$

$$= 0$$

$$\therefore S = 0 \text{ as } w \neq 0.$$

METHOD 2

$$S = \frac{w}{1-w^2} + \frac{w^2}{1-w^4} + \frac{w^3}{1-w} + \frac{w^4}{1-w^3}$$

$$= \frac{w}{1-w^2} + \frac{w^4}{w^2-w} + \frac{w^4}{w-w^2} + \frac{w}{w^2-1}$$

$$= \frac{w}{1-w^2} - \frac{w}{1-w^2} + \frac{w^4}{w^2-w} - \frac{w^4}{w^2-w}$$

$$= 0$$